Groundwater Management: Hydrogeology, Protection, and Governance

Hydrogeology and Aquifer Dynamics

Groundwater represents the primary freshwater storage on Earth, supplying drinking water to over 50 percent of the global human population and over 40 percent of agricultural irrigation. Aquifers are underground layers of water-bearing permeable rock, gravel, sand, or silt. Unconfined aquifers are replenished directly by surface infiltration, while confined aquifers are sandwiched between low-permeability aquitards.

Hydrogeological characterization relies on hydraulic conductivity, storativity, transmissivity, well pumping tests, piezometric monitoring networks, and isotopic tracer analysis to calculate safe yield extraction rates and model groundwater flow fields.

Over-Exploitation, Subsidence, and Saline Intrusion

Unregulated abstraction exceeding natural recharge causes severe water table declines, drying shallow wells, reducing river baseflows, and degrading groundwater-dependent wetlands. Excessive drawdown in coastal areas causes saltwater intrusion, pulling dense ocean saltwater into freshwater aquifers.

Land subsidence occurs when prolonged groundwater pumping reduces pore pressure, causing clay and silt beds within aquifers to consolidate irreversibly. Subsidence damages surface buildings, pipelines, roads, and flood protection levees in major agricultural basins and coastal megacities.

Managed Aquifer Recharge (MAR) and Artificial Replenishment

Managed Aquifer Recharge (MAR) encompasses engineered techniques designed to intentionally augment groundwater storage. Methods include surface infiltration basins, percolation ponds, aquifer storage and recovery (ASR) injection wells, and check dam structures built across ephemeral streams.

MAR projects capture surplus monsoon runoff, treated stormwater, or reclaimed municipal effluent, filtering water naturally through the vadose zone into underlying aquifers. MAR improves drought resilience, prevents seawater intrusion, and restores depleted groundwater storage.

Groundwater Quality Protection and Contaminant Mitigation

Groundwater contamination arises from point sources (leaking underground fuel tanks, industrial solvent spills, landfill leachate) and non-point sources (nitrate fertilizer leaching, pesticide runoff, municipal septic systems). Geogenic contaminants like geogenic arsenic and fluoride pose severe widespread health risks.

Protection strategies enforce wellhead protection zones (WHPZs), restrict chemical storage near recharge areas, mandate impermeable liners for landfills, and implement permeable reactive barriers (PRBs) or pump-and-treat remediation systems for contaminated plumes.